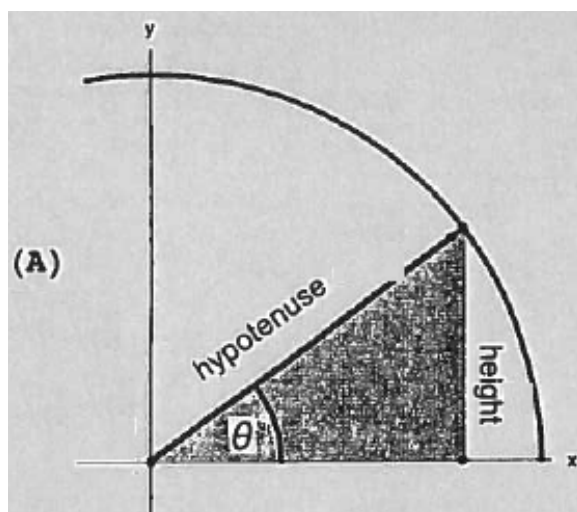


Problem 3

Problem 3

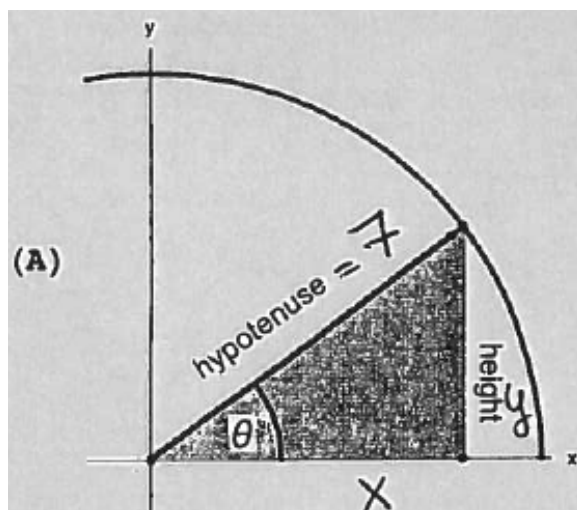
Problem 1 A part of a circle centered at the origin with radius $r = 7$ cm is given in the figure (A) below. A right triangle is formed in the first quadrant (see Figure (A)). One of its sides lies on the x -axis. Its hypotenuse runs from the origin to a point on the circle. The hypotenuse makes an angle θ with the x -axis. Assume that the angle θ changes at the rate $\frac{d\theta}{dt} = 0.2$ radians per second.



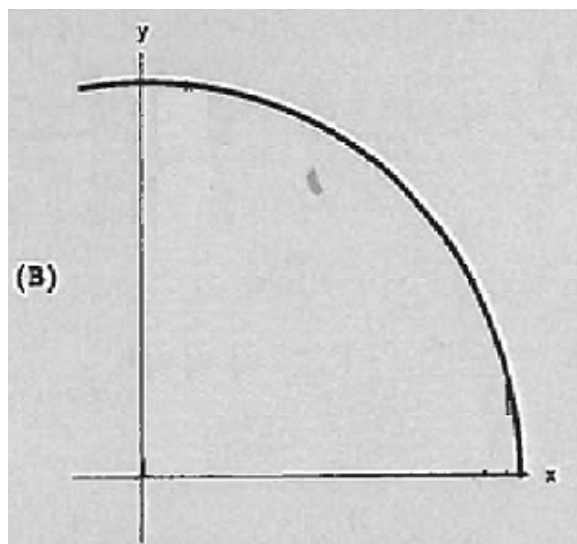
(a) Label Figure A.

Explanation.

Problem 3

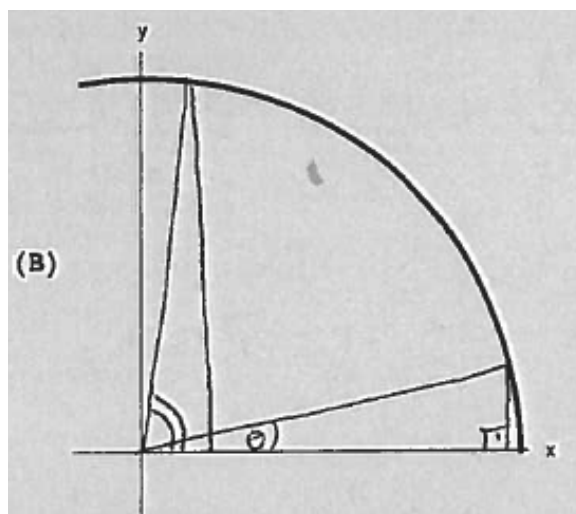


- (b) In Figure B, draw the triangle twice; once when θ is small and once more, when θ is close to $\frac{\pi}{2}$.



Explanation.

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- (c) Find the rate of change of the height of the triangle when $\theta = \frac{\pi}{3}$

Explanation. $y = 7 \sin \theta$

$$\begin{aligned} \frac{dy}{dt} &= \frac{dy}{d\theta} \cdot \frac{d\theta}{dt} \\ \frac{dy}{dt} &= 7 \cos \theta \cdot \frac{d\theta}{dt} \\ \left[\frac{dy}{dt} \right]_{\theta=\frac{\pi}{3}} &= 7(1/2)(0.2) \\ &= 0.7 \text{ cm/sec} \end{aligned}$$

- (d) Find the rate of change of the area of the triangle when $\theta = \frac{\pi}{3}$

Explanation. First, we need to find x : $x = 7 \cos \theta$. From part c, we

Problem 3

know $y = 7 \sin \theta$

$$A = \frac{1}{2}xy$$

$$A = \frac{49}{2} \cos(\theta) \sin(\theta)$$

$$\frac{dA}{dt} = \frac{49}{2}(-\sin^2(\theta) + \cos^2(\theta)) \cdot \frac{d\theta}{dt}$$

$$\left[\frac{dA}{dt} \right]_{\theta=\frac{\pi}{3}} = \frac{49}{2} \left(- \left(\frac{\sqrt{3}}{2} \right)^2 + \left(\frac{1}{2} \right)^2 \right) \cdot 0.2$$

$$= \frac{49}{2} \cdot \frac{-1}{2} \cdot 0.2$$

$$= \frac{-4.9}{2} \text{ cm}^2/\text{sec}$$